Exercise Week 6

Introduction to Econometrics

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Instructions: This exercise is based on the paper Ginsburgh and van Ours (2007), Expert Opinion and Musical Competition, American Economic Review, 93(1), 289-296. After reading the introduction of the paper as well as section I and II you can replicate the paper and conduct a full blown IV analyzes (with and without reading the rest of the paper.) by answering the following questions:

Please note concerning how to produce nice tables in STATA: estpost, esttab and eststo are commands that enables you to save summary statistics and regression results directly as a Word, LaTex or ASCII format. estpost, esttab and eststo are not standard commands included in Stata15, but available in the estout package. This is a user written package that can be downloaded to, and used from, your local computer.

1. What is the main research question of the paper?

Solution:

- Do experts' opinion reflect true quality, or do they influence economic outcomes independently of their value as a signal of quality?
- Example of a **valid** instrument
- Many economic decisions made by "experts"
  - Financial analysts
  - Academic committees
  - Olympic juries

2. What are the main methodological challenges in order to answer this research questions? What is the main idea of the authors to solve it. Answer verbally and formally.

Solution:

- Question: do experts' opinion reflect true quality, or do they influence economic outcomes independently of their value as a signal of quality?
- Why not regress measure of success on expert opinions using OLS?
  - True quality is likely to drive success
  - True quality unobservable and likely to be correlated with experts' opinions
  - Ommitted variable bias
- Solution: use an instrument that is
  - Correlated with ranking (relevant)
  - Uncorrelated with unobservable quality (valid)
- Ginsburgh and Van Ours look at Queen Elizabeth musical competition
  - Held every year in Belgium
  - Best known international competition for piano and violin
- Unusual characteristic: finalists get a single week to study a completely new concerto
  - Order in which finalists play is randomized
  - Two finalists perform per evening
- Instrument: order in which finalists play
  - Order is random: uncorrelated with quality (**valid** instrument)
  - Order may be correlated with ranking in competition (**relevant** instrument)
- In terms of our general notation:
  - $y_i =$ measures of success: presence in catalogues and ratings by Belgian music critics
  - $z_i =$ ranking in competition ($:= r_i$ – endogenous)
  - $\mathbf{x}_i =$ order of finalists (exclusion restriction)
  - $q_i =$ quality of finalists, unobserved
- Base equation:

$$\begin{aligned} y_i &= \delta_0 + \delta_1 r_i + \delta_2 q_i + u_i \\ &= \delta_0 + \delta_1 r_i + \varepsilon_i \end{aligned}$$

  where $\varepsilon_i = \delta_2 q_i + u_i$ (q_i not observed)
- r_i endogenous if

- $\delta_2 \neq 0$ (quality matters for success)
- $\text{cov}\left(r_i, q_i\right) \neq 0$ (ranking correlated with quality)

- $\mathbf{x}_i$: order of appearance exogenous
  - $\mathbb{E}\left(\mathbf{x}_i \epsilon_i\right) = 0$

3. Please produce descriptive statistics of the data.

Solution:

female	0.258
	(0.439)
age	24.53
	(3.507)
ussr	0.159
	(0.367)
usa	0.227
	(0.421)
belgium	0.0833
	(0.277)
gccd	6.485
	(19.71)
bll	10.85
	(22.96)
numrec	17.33
	(42.05)
presen	1.644
	(1.160)
first	0.0833
	(0.277)
odd	0.500
	(0.502)
late	0.500
	(0.502)
ranking	6.500
	(3.465)
N	132

mean coefficients; sd in parentheses

4. Run OLS regressions of number of critics on the variable negranking. How would you interpret the result?

Solution:

5. Estimated equation:

$$y_i = \delta_0 + \delta_1 r_i + \varepsilon_i$$

	(1)
	b/se
negranking	1.475
	0.274
_cons	23.229
	2.086
r2	0.180

How ranking linearly predicts the success criterium number of critics' review. **One higher rank is associated linearly with approximately 1.5 more critics. This association is significant as can be seen by the small standard error in relation to b. $\frac{1.475}{0.274}$ generates a large t-value that is much larger than 2.**

6. Run first stage regressions for two different sets of instruments and test for relevance of the instruments.

Solution:

- First stage:

$$r_i = \beta_0 + \beta_1 first_i + \beta_2 female_i + \beta_3 late_i + \nu_i$$

where $first_i = \mathbb{1}\{i$ was first to perform in competition$\}$;
$female_i = \mathbb{1}\{i$ is female$\}$;
$late_i = \mathbb{1}\{i$ was second to play on particular evening$\}$

- Test for relevance: test for joint significance of instruments

```
. gen negranking  = -ranking

. reg negranking first female late, robust

Linear regression                               Number of obs =     132
                                                F(  3,   128) =    8.19
                                                Prob > F      =  0.0000
                                                R-squared     =  0.1478
                                                Root MSE      =  3.2362
```

negranking	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
first	-2.958409	.9146216	-3.23	0.002	-4.768144	-1.148674
female	-1.856241	.6418419	-2.89	0.004	-3.126235	-.5862471
late	1.130227	.6097634	1.85	0.066	-.0762945	2.336748
_cons	-6.340457	.511137	-12.40	0.000	-7.351828	-5.329085

First stage:

$$r_i = \beta_0 + \beta_1 first_i + \nu_i$$

```
. reg negranking first, robust

Linear regression                               Number of obs =     132
                                                F(  1,   130) =   24.82
                                                Prob > F      =  0.0000
                                                R-squared     =  0.0750
                                                Root MSE      =  3.3454
```

negranking	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
first	-3.421488	.6867483	-4.98	0.000	-4.780137	-2.062838
_cons	-6.214876	.3127542	-19.87	0.000	-6.833623	-5.596129

- $first$ is highly relevant
  - But joint test for significance instruments: F-statistic smaller than 10
- Musicians who play during first evening rank 3 places lower on average
- Females are ranked 2 places lower on average
- Those who perform second gain 1 rank
- Though the order of performing is random, it does affect the final ranking
  - Jury hears concerto for first time too – maybe they get less severe as competition unfolds

7. Conduct a 2SLS estimation for the two sets of instruments using number of critics as outcome variable. What is your conclusion?

Solution:

Second stage – success measured by critics' reviews

$$y_i = \delta_0 + \delta_1 \widehat{r}_i + \varepsilon_i$$

One instrument:

```
.        ivregress 2sls critics (negranking = first), robust

Instrumental variables (2SLS) regression               Number of obs =     132
                                                       Wald chi2(1)  =    9.00
                                                       Prob > chi2   =  0.0027
                                                       R-squared     =  0.1206
                                                       Root MSE      =  11.254

------------------------------------------------------------------------------
             |               Robust
     critics |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
  negranking |   2.321256   .7736948     3.00   0.003      .804842     3.83767
       _cons |    28.7321   5.308896     5.41   0.000     18.32686    39.13735
------------------------------------------------------------------------------
Instrumented:  negranking
Instruments:   first
```

(OLS estimate 1.475, SE 0.274)

Multiple instruments:

```
.        ivregress 2sls critics (negranking = first female late), robust

Instrumental variables (2SLS) regression               Number of obs =     132
                                                       Wald chi2(1)  =   12.50
                                                       Prob > chi2   =  0.0004
                                                       R-squared     =  0.1165
                                                       Root MSE      =   11.28

------------------------------------------------------------------------------
             |               Robust
     critics |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
  negranking |   2.350242   .6647744     3.54   0.000     1.047308    3.653176
       _cons |   28.92051   4.580049     6.31   0.000     19.94378    37.89724
------------------------------------------------------------------------------
Instrumented:  negranking
Instruments:   first female late
```

(OLS estimate 1.475, SE 0.274)

- Effect of ranking on success is *larger* for 2SLS than OLS. **One rank higher leads to 2.3 more critics. This effect is significantly different from zero, as can be seen by the p-value of that is smaller than 0.05**.
  - Ranking and true quality negatively correlated?
  - Expert judges have different opinion than critics and record buyers.
- Expert opinion has effect on success over and above signaling
- It pays to do well in competitions
- But: results from competitions are affected by unexpected factors
  - Sheds doubt on ability of experts to cast objective judgments

8. Conduct postestimation test after 2SLS concerning the relevance of the instrument, overidentifying restrictions and the endogeneity of regressor. What is your conclusion? You can use the postestimation commands `estat endogenous`, `estat overid`, `estat firststage`.

Solution:

- Relevance of instruments

```
.       estat firststage

First-stage regression summary statistics
```

Variable	R-sq.	Adjusted R-sq.	Partial R-sq.	Robust F(3,128)	Prob > F
negranking	0.1478	0.1278	0.1478	8.18832	0.0000

Using all three variables as instruments leads to the weak instrument problem: In this case the small sample bias of the 2SLS estimator can be large; and small violations of instrument validity can yield large asymptotic bias in the 2SLS estimator.

- Overidentifying restrictions (Hansen/Sargan's test)

```
.       estat overid

Test of overidentifying restrictions:

Score chi2(2)         =  1.28449  (p = 0.5261)
```

Specification test, testing all the restriction of the model. Only if confident that all the other assumptions are fulfilled, large J evidence for endogeneity of some of the instruments. H0: $\mathbb{E}(\mathbf{x}_i\varepsilon_i) = \mathbf{0}$ Here we cannot reject the HO, that the orthogonality assumptions are valid.

- Endogeneity of regressor (Hausman test) under the maintained assumption of instrument validity.

```
.       estat endogenous

Tests of endogeneity
Ho: variables are exogenous

Robust score chi2(1)          =  1.99995  (p = 0.1573)
Robust regression F(1,129)    =  2.22072  (p = 0.1386)
```

1. H_0: plim $\left(\widehat{\boldsymbol{\beta}}_{\text{2sls}} - \widehat{\boldsymbol{\beta}}_{\text{OLS}}\right) = \mathbf{0}$ ($ranking$ is exogenous) vs. H_a: plim $\left(\widehat{\boldsymbol{\beta}}_{\text{2sls}} - \widehat{\boldsymbol{\beta}}_{\text{OLS}}\right) \neq \mathbf{0}$ ($ranking$ not exogenous)
2. Value of F-statistic= 2.22
3. P-value = 0.13
4. Conclusion: do not reject the null hypothesis, insufficient evidence to conclude that $ranking$ is endogenous. But as IV is still consistent paper continues using IV. Not uncommon.

9. Now conduct a 2SLS in 2 separate steps, together with the Hausman test.

Solution:

1. Estimate first stage by OLS, obtain residuals

2. Add first stage residuals to second stage regression, estimate by OLS:

```
. reg critics negranking resid1st, robust

Linear regression                               Number of obs =     132
                                                F(  2,   129) =   17.29
                                                Prob > F      =  0.0000
                                                R-squared     =  0.1909
                                                Root MSE      =  10.919

------------------------------------------------------------------------------
             |               Robust
     critics |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
  negranking |   2.350242   .6051227     3.88   0.000     1.152992    3.547492
    resid1st |   -1.02749   .6894938    -1.49   0.139     -2.39167    .3366905
       _cons |   28.92051    4.24448     6.81   0.000      20.5227    37.31832
------------------------------------------------------------------------------
```

- t-test on `resid1st` constitutes Hausman test
  - We fail to reject the null hypothesis that ranking is exogenous